

Diagnosis and Management of Resistant Hypertension

A Review

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IMPORTANCE Hypertension, defined as office systolic blood pressure (SBP) 130 mm Hg or greater and/or diastolic blood pressure 80 mm Hg or greater, affects 43.9% of women and 49.5% of men in the US. Approximately 19.7% of patients treated for hypertension have apparent resistant hypertension (blood pressure $\geq$ 130/80 mm Hg) despite using 3 or more antihypertensive medications, preferably a renin-angiotensin system blocker, a calcium channel blocker, and a thiazide-type diuretic, at maximally tolerated doses.

OBSERVATIONS Approximately 10% of patients treated for hypertension have true resistant hypertension confirmed with home or 24-hour ambulatory blood pressure monitoring to exclude white-coat hypertension (approximately 37.5% of apparent resistant hypertension) and after excluding medication nonadherence (approximately 50%) and secondary hypertension such as primary aldosteronism (approximately 5%-25%). Conditions associated with resistant hypertension include obesity, diabetes, chronic kidney disease, and sleep apnea. Resistant hypertension is associated with increased risk of cardiovascular death vs controlled blood pressure at 5 years to 10 years (absolute risk increase, 10.3% [95% CI, 8.7%-12.1%]). Lifestyle modifications for resistant hypertension include a low-sodium diet (<1500 mg/d), reducing or avoiding alcohol, 150 min/wk or more of aerobic exercise, and weight loss. Illicit drugs (eg, cocaine) and medications that increase blood pressure (eg, nonsteroid anti-inflammatory drugs, serotonin-norepinephrine reuptake inhibitors) should be avoided. Sleep apnea should be treated when diagnosis is confirmed. Pharmacologic optimization includes use of combination tablets of antihypertensives; intensifying diuretic therapy by using chlorthalidone; and sequential addition of antihypertensive medications using evidence-based algorithms. In a meta-analysis of 20 studies (9 randomized clinical trials [RCTs] and 11 observational studies [331 participants]), use of antihypertensive therapies that combine 2 to 3 medications into a single formulation reduced SBP by -3.99 mm Hg (95% CI, -7.92 to -0.07) vs equivalent doses given separately. For patients with apparent or true resistant hypertension who have an estimated glomerular filtration rate of 45 mL/min/1.73 m² or greater and a serum potassium level of 4.5 mmol/L or less, adding spironolactone (25-50 mg/d) compared with placebo lowers office SBP by -13.3 mm Hg (95% CI, -17.89 to -8.72 [4 RCTs]) and 24-hour ambulatory SBP by -8.46 mm Hg (95% CI, -12.54 to -4.38 [2 RCTs]) in a network meta-analysis of 24 RCTs (3485 patients with resistant hypertension). A meta-analysis of 10 RCTs (2478 participants) reported that compared with a sham procedure, catheter-based renal denervation, which disrupts the sympathetic nerves in the renal artery walls, decreased 24-hour ambulatory SBP by -4.4 mm Hg (95% CI, -6.1 to -2.7) and office SBP by -6.6 mm Hg (95% CI, -9.7 to -3.6).

CONCLUSIONS AND RELEVANCE True resistant hypertension affects 10% of patients treated for hypertension and is diagnosed after excluding white-coat hypertension, medication nonadherence, and secondary hypertension such as primary aldosteronism. First-line treatment includes lifestyle modifications, diuretic therapy with chlorthalidone, and combination tablets of antihypertensives. Spironolactone and renal denervation decrease blood pressure in patients with true resistant hypertension.

JAMA. doi:10.1001/jama.2026.1221
Published online March 23, 2026.

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Hypertension, defined as systolic blood pressure (SBP) 140 or greater and/or diastolic blood pressure (DBP) 90 mm Hg or greater or use of medication for hypertension, affects 25% to 30% of adults worldwide.¹ When defined by blood pressure of 130/80 mm Hg or greater or receiving antihypertensive therapy, the 2017-2020 National Health and Nutrition Examination Survey of US adults aged 18 to 80 years reported the prevalence of hypertension as 49.5% (59.0 million) for men and 43.9% (56.3 million) for women.² Approximately 19.7% of adults taking antihypertensive medications in the US meet the 2025 American Heart Association/American College of Cardiology (AHA/ACC) diagnosis of apparent resistant hypertension, defined as an office SBP of 130 mm Hg or greater and/or DBP of 80 mm Hg or greater despite use of 3 medications (preferably an angiotensin-converting enzyme [ACE] inhibitor or an angiotensin receptor blocker, a calcium channel blocker, and a thiazide-type diuretic) at maximally tolerated doses.³ True resistant hypertension is confirmed after exclusion of white-coat hypertension, nonadherence to antihypertensive medications, and secondary causes of hypertension. This review focuses on the epidemiology, pathophysiology, diagnosis, and treatment of apparent and true resistant hypertension (Box 1).

Methods

We searched PubMed (MEDLINE) for English-language articles from January 2015 through January 2026 using the terms *resistant hypertension* or *refractory hypertension*. We focused on studies published from 2015 to 2026 to provide evidence aligned with contemporary diagnostic thresholds and guideline definitions but included earlier studies when clinically relevant. We prioritized evidence from randomized clinical trials (RCTs), meta-analyses, and major guideline statements. Reference lists from relevant articles and guidelines were searched for additional pertinent studies. Of 214 manuscripts retrieved from the search, 87 were included in this review, consisting of 25 RCTs, 17 meta-analyses, 5 longitudinal prospective observational studies, 9 cross-sectional studies, 17 guidelines, and 14 pathophysiological reviews.

Epidemiology and Risk Factors

In 2 separate cohort studies (800 and 2316 participants) of patients with hypertension, 19.4% of participants with initially controlled blood pressure with antihypertensive medications progressed to apparent resistant hypertension (blood pressure $\geq$ 130/80 mm Hg) over 9 years.⁴ In a systematic review and meta-analysis of patients taking antihypertensive medications (91 cross-sectional and cohort studies [3 207 911 participants]), the prevalence of true resistant hypertension at a higher threshold of blood pressure of 140/90 mm Hg or greater was 10.3% (95% CI, 7.6%-13.2%).⁵ Higher prevalence of apparent resistant hypertension (up to 50%) and true resistant hypertension (22.9%) is reported among patients with chronic kidney disease (CKD), particularly in those with albuminuria.^{5,6}

In a cross-sectional study of 37 061 US patients with apparent resistant hypertension, risk factors associated with resistant hypertension were CKD (odds ratio [OR], 1.84 [95% CI, 1.78-1.90]), Black race (OR, 1.68 [95% CI, 1.62-1.75]), diabetes (OR, 1.58 [95% CI, 1.53-1.63]), obesity (OR, 1.46 [95% CI, 1.42-1.51]), cardiovascular disease

Box 1. Common Questions About Resistant Hypertension

How is true resistant hypertension defined and distinguished from apparent resistant hypertension?

Resistant hypertension is defined as systolic blood pressure 130 mm Hg or greater and/or diastolic blood pressure 80 mm Hg or greater despite the use of 3 or more antihypertensive medications of different classes at maximal tolerated doses (preferably including an angiotensin-converting enzyme inhibitor or angiotensin receptor blocker, a calcium channel blocker, and a thiazide-type diuretic), after excluding white-coat hypertension, nonadherence to antihypertensive medications, and secondary hypertension (such as primary aldosteronism).

What are the common risk factors or causes of resistant hypertension?

Volume overload due to high sodium intake, primary aldosteronism, inappropriate and renin-independent aldosterone excess without overt primary aldosteronism, obesity, obstructive sleep apnea, and chronic kidney disease are common among patients with resistant hypertension. Nonsteroidal anti-inflammatory drugs, some antidepressants (such as serotonin-norepinephrine reuptake inhibitors), and illicit drugs (such as cocaine) are associated with poor blood pressure control.

How should true resistant hypertension be treated?

Management of true resistant hypertension includes lifestyle modifications (adherence to a low sodium diet [$<$ 1500 mg/d], reducing or avoiding alcohol, moderate-intensity exercise [$\geq$ 150 min/wk of regular, moderate intensity aerobic exercise, and $\geq$ 2 d/wk of resistance exercise], and weight loss). For patients with resistant hypertension, the use of combinations pills, intensifying diuretic therapy by using preferably chlorthalidone (especially in patients with chronic kidney disease), and the addition of a mineralocorticoid receptor antagonist (eg, spironolactone) are recommended for patients with estimated glomerular filtration rate of 45 mL/min/1.73 m² or greater and serum potassium level of 4.5 mmol/L or less. Patients who experience adverse effects with spironolactone can be switched to amiloride or eplerenone. Other therapeutic options include β -blockers, α -adrenergic blockers, centrally acting antihypertensive agents, nondihydropyridine calcium channel blockers, direct vasodilators, or a dual endothelin receptor antagonist (eg, apocritentan), which may be also used as alternatives to spironolactone in case of adverse events or contraindication or added sequentially to a regimen that includes spironolactone.

(OR, 1.34 [95% CI, 1.30-1.39]), older age (OR, 1.11 [95% CI, 1.10-1.11] for every 5-year increase over age of 17 years), and male sex (OR, 1.06 [95% CI, 1.03-1.10]).⁷ A systematic review and meta-analysis (6 observational studies [1465 participants]) reported an association between obstructive sleep apnea (OSA) and resistant hypertension (pooled OR, 2.84 [95% CI, 1.70-3.98]; absolute rates not reported).⁸ A meta-analysis of 23 cohort studies with more than 600 000 participants reported an association between daily alcohol consumption and the presence of hypertension with risk ratios of 0.89 (95% CI, 0.84-0.94) for 0 g/d of alcohol, 1.11 (95% CI, 1.07-1.15) for 24 g/d, 1.22 (95% CI, 1.14-1.30) for 36 g/d, and 1.33 (95% CI, 1.18-1.49) for 48 g/d, compared with participants who drank 12 g/d of alcohol.^{9,10} Obesity (body mass index $\geq$ 30, calculated as weight in kilograms divided by square of height in meters) was present in 29 560 of 60 327 patients (49%) with resistant hypertension in a 2013 US observational study.⁷

Pathophysiology

The pathophysiology of resistant hypertension involves the complex interplay of several neurohormonal pathways, including inappropriate and renin-independent aldosterone excess¹¹; sympathetic,¹² endothelin-1,¹³ and vasopressin overactivity¹⁴; impaired natriuretic peptide system activity¹⁵; and impaired endothelial function.¹⁶ These factors contribute to volume and sodium overload, increases in arterial stiffness, and kidney fibrosis. Excess dietary sodium intake (≥ 2300 mg/d) causes low plasma renin activity and promotes volume overload, which diminishes the effectiveness of renin-angiotensin system (RAS) blockers and increases the requirement for diuretics at higher doses or in combination to achieve adequate natriuresis.^{17,18} Low dietary potassium intake (<1500 mg/d) may also contribute to elevated blood pressure through mechanisms such as increased sodium retention by the kidneys and increased vascular tone.^{19,20} Obstructive sleep apnea may independently increase blood pressure through sympathetic overdrive, renin-angiotensin-aldosterone system activation, endothelial dysfunction, and possible fluid retention.^{21,22}

Clinical Presentation

Office Blood Pressure Measurements

Patients with apparent resistant hypertension take 3 or more antihypertensive medications and have elevated blood pressure ($\geq 130/80$ mm Hg), typically detected in the office.^{3,23,24} While the 2025 AHA/ACC guidelines use a threshold of office SBP of 130 mm Hg or greater and/or of DBP 80 mm Hg or greater to define resistant hypertension,³ the 2023 European Society of Hypertension and the 2024 European Society of Cardiology guidelines define resistant hypertension with an office blood pressure of 140/90 mm Hg or greater.^{23,24} Office blood pressure measurements should be obtained with the patient in the seated position for at least 5 minutes using a proper technique with a [validated](#) oscillometric electronic device and a correct cuff size^{3,23,24} and recorded as the average of at least 2 measurements taken at least 1 minute apart. Patients with apparent resistant hypertension based on office blood pressure should undergo out-of-office blood pressure measurement with either home blood pressure monitoring or 24-hour ambulatory blood pressure monitoring (ABPM) to exclude white-coat effect.³

Out-of-Office Blood Pressure Measurements

Resistant hypertension with home blood pressure monitoring is defined as an average of 130/80 mm Hg or greater for all recorded blood pressures, which corresponds to the office blood pressure threshold of 130/80 mm Hg or greater.^{3,23,24} Self-monitoring of blood pressure at home requires that patients use a validated automated device with a correct cuff size matched to the arm size and receive instructions from a clinician or a trained nurse. Measurements should be taken twice daily for a minimum of 3 days, but preferably over 1 week, prior to antihypertensive medication intake.^{3,23} The patient should be in the seated position for 5 minutes of rest and then should record 2 blood pressure measurements taken 1 minute apart.

Out-of-office measures can also be obtained with ABPM, which is performed with a small fully automated portable blood pressure

monitor worn on the upper arm for 24 hours. Over the 24-hour cycle, ABPM provides blood pressure measurements every 15 to 30 minutes during daytime and at less frequent intervals during nighttime (30-60 minutes). The 24-hour ambulatory blood pressure threshold for resistant hypertension is an average of 125/75 mm Hg or greater in the US guidelines and 130/80 mm Hg or greater in the European guidelines.^{23,24}

Differentiating Apparent and True Resistant Hypertension

The differential diagnosis for apparent resistant hypertension includes white-coat hypertension, defined as office blood pressure 130/80 mm Hg or greater, with home blood pressure less than 130/80 mm Hg or 24-hour ABPM less than 125/75 mm Hg, and medication nonadherence ([Figure](#)).^{3,23,24} In an observational cohort study from Spain, 3110 of 8295 patients (37.5%) with apparent resistant hypertension (office blood pressure $\geq 140/90$ mm Hg) had 24-hour ambulatory blood pressure less than 130/80 mm Hg.²⁵ In other studies, nonadherence has been reported in up to 50% of patients.²⁶ Assessment of adherence should include a nonjudgmental discussion with patients and review of pharmacy refill data. In some cases, if uncertainty persists, direct measurement of antihypertensive drug or metabolite levels in urine or blood can be performed, typically using liquid chromatography-tandem mass spectrometry.²⁶

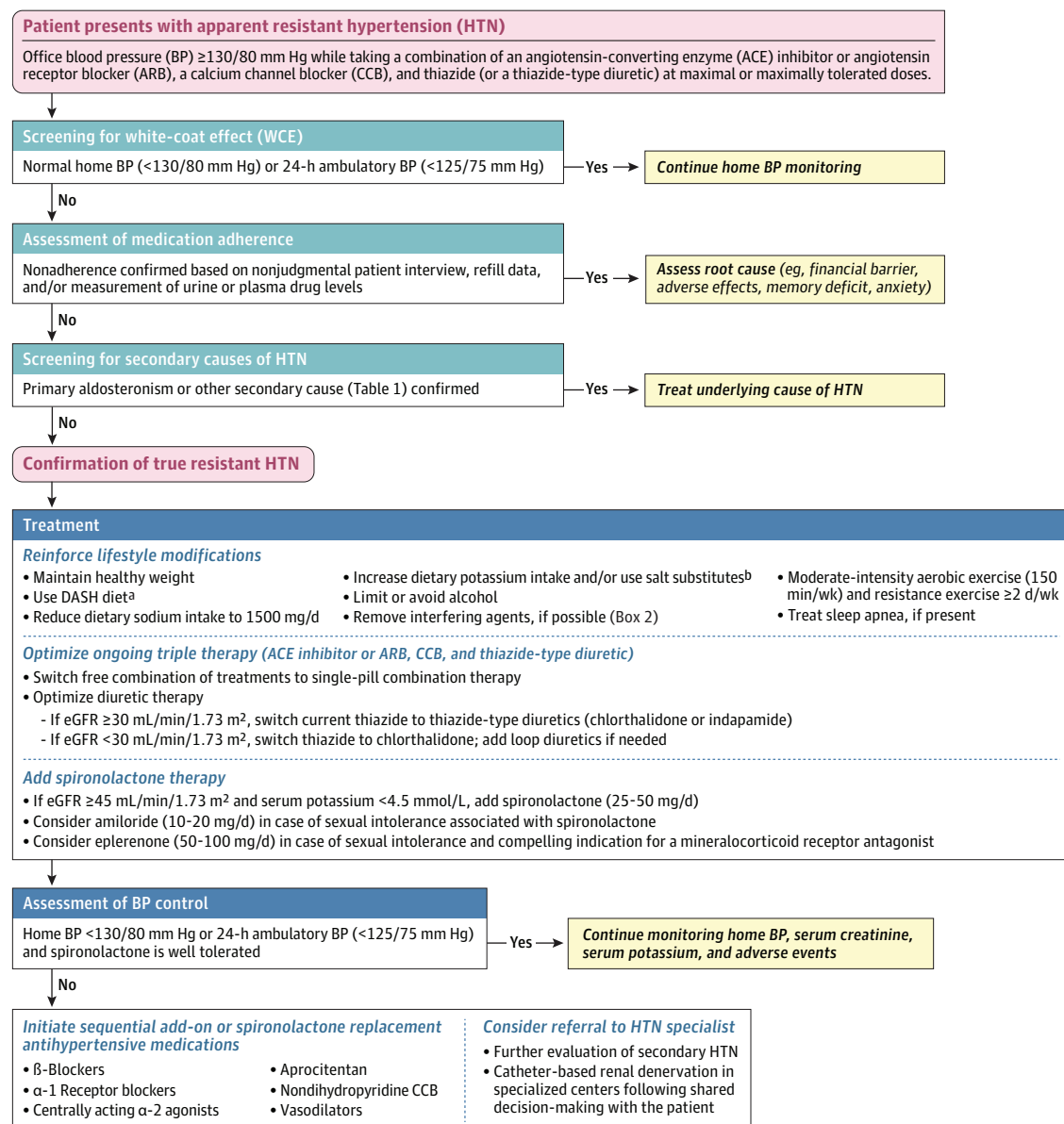
Secondary Hypertension

Secondary hypertension is the presence of hypertension attributable to identifiable and often treatable underlying conditions ([Table 1](#)).^{3,23,24} The prevalence of secondary hypertension is higher among patients with resistant hypertension ($\approx 25\%$) than among hypertensive individuals without resistant hypertension ($\approx 10\%$) ([Table 1](#)).^{3,27,28} All patients with resistant hypertension should be screened for secondary causes of hypertension, particularly primary aldosteronism, which is characterized by high plasma aldosterone levels and low plasma renin levels when measured simultaneously, leading to an elevated aldosterone-to-renin ratio.^{3,23,24} Primary aldosteronism is the most common secondary cause of resistant hypertension (range, 5%-25%).^{3,23,24,27-29}

Medication Review

Patients with apparent resistant hypertension should also have a medication review to identify medications that may increase blood pressure directly or counteract the blood pressure-lowering effect of antihypertensive treatments ([Box 2](#)).³⁰ In a US cross-sectional survey (National Health and Nutrition Examination Survey) including 27 599 individuals, 18.5% of adults 18 years or older with hypertension reported using medications that may increase blood pressure, including antidepressants (8.7%), nonsteroidal anti-inflammatory drugs (6.5%), steroids (1.9%), and oral estrogens (1.7%).³¹ Whenever feasible, all medications that may increase blood pressure should be discontinued or replaced with alternatives that do not affect blood pressure, because even modest drug-induced blood pressure elevations (2-5 mm Hg) may increase cardiovascular risk in patients with resistant hypertension. However, decisions about medication withdrawal should be individualized based on the indication and clinical benefit of the drug, which may outweigh a modest increase in blood pressure, the availability of alternative medications, and patient preferences.

Figure. Algorithm for Management of Resistant Hypertension

^aDASH indicates Dietary Approaches to Stop Hypertension (<https://www.nhlbi.nih.gov/education/dash-eating-plan>).^bExcept in patients with estimated glomerular filtration rate (eGFR) less than 45 mL/min/1.73 m².

Management of True Resistant Hypertension

Management of true resistant hypertension involves intensifying lifestyle modifications, and stepwise addition of antihypertensive medications targeting different pathophysiological mechanisms (Figure). Using evidence-based treatment algorithms,^{3,23,24,27} treatment should be individualized, based on patient preferences, comorbid conditions, kidney function, potential medication adverse effects, and drug interactions. Treatment focuses on mean blood pressure reduction as the primary target, because absolute blood pressure lowering is an accepted surrogate marker to reduce cardiovascular morbidity and mortality.³²⁻³⁴ For example, a meta-analysis of 48 randomized trials (344 716 participants; median follow-up, 4.15 years) reported that a 5-mm Hg decrease in office SBP was associated with

decreased incidence of major adverse cardiovascular events among participants with both prior cardiovascular disease (absolute risk reduction, -1.6% [95% CI, -1.9% to -1.2%]; hazard ratio [HR], 0.89 [95% CI, 0.86-0.92]) and without prior cardiovascular disease (absolute risk reduction, -2.2% [95% CI, -2.5% to -1.9%]; HR, 0.91 [95% CI, 0.89-0.94]).^{32,33}

Lifestyle Modifications

Patients with resistant hypertension can decrease blood pressure by reducing sodium and alcohol intake, increasing dietary potassium intake, or using salt substitutes (except in those with estimated glomerular filtration rate [eGFR] <45 mL/min/1.73 m²), using the **Dietary Approaches to Stop Hypertension** eating plan and en-

Table 1. Common and Uncommon Causes of Secondary Hypertension^a

Secondary causes of hypertension	Prevalence, %	Indications for screening
Common causes		
Obstructive sleep apnea	25-50	Snoring, gasping during sleep; daytime sleepiness; resistant hypertension
Chronic kidney disease	14	Diabetes, obstruction, hematuria; urinary frequency and nocturia; urinary incontinence, analgesic abuse; family history of polycystic kidney disease; elevated serum creatinine level; abnormal urinalysis
Primary aldosteronism	5-25	Resistant hypertension; hypertension with hypokalemia (spontaneous or diuretic induced); hypertension and muscle cramps or weakness; hypertension and incidentally discovered adrenal mass; hypertension and obstructive sleep apnea; hypertension and family history of early-onset hypertension or stroke
Drug- or alcohol-induced	2-20	See Box 2 for details
Renovascular hypertension	0.1-5	Resistant hypertension; hypertension with abrupt onset or worsening or increasingly difficult to control; generalized atherosclerotic disease (coronary or peripheral artery disease; carotid, abdominal, or femoral bruit; abdominal systolic diastolic bruit; flash pulmonary edema [atherosclerotic renal artery stenosis]); acute deterioration of renal function (eGFR decrease of 30% or greater) after RAS blocker initiation, recent renal insufficiency, small unilateral kidney; abdominal or lumbar bruit; early-onset hypertension, especially in women (renal artery stenosis due to fibromuscular dysplasia)
Uncommon causes		
Hypothyroidism	<1	Dry skin; cold intolerance; constipation; hoarseness; weight gain
Hyperthyroidism	<1	Warm, moist skin; heat intolerance; nervousness; tremulousness; palpitations, insomnia; weight loss; eye signs (upper eyelid retraction, Graves orbitopathy); diarrhea; proximal muscle weakness; goiter, thyroid nodule
Pheochromocytoma/paraganglioma	<0.6	Resistant hypertension; paroxysmal hypertension or crisis superimposed on sustained hypertension; blood pressure lability, headache, sweating, palpitations, piloerection; positive family history of pheochromocytoma/paraganglioma or presence of associated genetic diseases (multiple endocrine neoplasia type 2 syndrome, von Hippel-Lindau, type 1 neurofibromatosis); adrenal incidentaloma
Aortic coarctation (undiagnosed or repaired)	0.1	Young adult with hypertension (age <30 y), diminished femoral pulses; blood pressure higher in upper extremities than in lower extremities; absent femoral pulses; continuous murmur over patient's back or chest or abdominal bruit; left thoracotomy scar (postoperative)
Cushing syndrome	<0.1	Rapid weight gain with central distribution; proximal muscle weakness; easy bruising; facial plethora; depression; hyperglycemia

Abbreviations: CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate; RAS, renin-angiotensin system.

^a Adapted from the 2025 American Heart Association guidelines.³

gaging in regular exercise.^{3,19} The AHA recommends sodium intake of less than 1500 mg/d for most adults with hypertension.³ In a meta-analysis (21 RCTs [31 949 participants]), use of salt substitutes that replace approximately 25% of sodium chloride with potassium chloride, compared with regular salt, was associated with lower SBP (−4.61 mm Hg [95% CI, −6.07 to −3.14]) and lower DBP (−1.61 mm Hg [95% CI, −2.42 to −0.79]).³⁵

A meta-analysis of 25 RCTs with 4874 participants reported that SBP decreased by −1.05 mm Hg (95% CI, −1.43 to −0.66) per kilogram of weight loss.³⁶ An RCT (140 patients with resistant hypertension) reported that a 4-month center-based program that included dietary counseling, weight management, and exercise reduced 24-hour ambulatory SBP at the end of the intervention by −7.0 mm Hg (95% CI, −8.5 to −4.0), compared with no change after a single counseling session (−0.3 mm Hg [95% CI, −4.0 to 3.4]).³⁷ Similarly, a 12-week aerobic exercise training program composed of 3 supervised training sessions per week including a 10-minute warm-up, 40 minutes of aerobic exercise (cycling and/or walking at 50% to 70% of maximum oxygen consumption and a 10-minute cooldown) resulted in reductions in 24-hour ambulatory SBP by −7.1 mm Hg (95% CI, −12.8 to −1.4) compared with usual care in a single-blinded RCT in patients with resistant hypertension.³⁸ The 2025 AHA guideline recommends 150 minutes or more per week of regular, moderate-intensity aerobic exercise and 2 or more days per week of resistance exercise for all patients with hypertension.³

In a meta-analysis of 36 RCTs with 2865 participants, among individuals consuming 2 or fewer alcoholic drinks per day, reductions

in alcohol intake were not associated with significant blood pressure changes. In contrast, among those consuming more than 2 drinks per day, reduced alcohol intake was associated with greater blood pressure reductions. The greatest reductions were observed among participants consuming 6 or more drinks per day who reduced intake by approximately 50%, with mean decreases in SBP of −5.50 mm Hg (95% CI, −6.70 to −4.30) and DBP of −3.97 mm Hg (95% CI, −4.70 to −3.25).³⁹

The 2025 ACC/AHA guidelines³ recommend screening patients with resistant hypertension for OSA with validated instruments such as the STOP-Bang Questionnaire.⁴⁰ For patients with resistant hypertension and OSA, continuous positive airway pressure therapy reduced 24-hour ambulatory SBP by an average of −5.92 mm Hg (95% CI, −8.72 to −3.11) vs control in a meta-analysis of 12 RCTs and 718 participants.⁴¹

Pharmacologic Therapy

The preferred medication combinations to treat apparent resistant hypertension are an ACE inhibitor or an angiotensin receptor blocker; a calcium channel blocker (typically a long-acting dihydropyridine); and a thiazide-type diuretic, preferably using single-pill combination formulations when possible (Table 2).^{3,23,24,27} A systematic review and meta-analysis (9 RCTs and 11 observational studies [331 participants]) reported that single-pill formulations that combined 2 to 3 medications decreased SBP by −3.99 mm Hg (95% CI, −7.92 to −0.07) compared with equivalent doses given as separate formulations.⁴²

Box 2. Medications and Substances Increasing Blood Pressure**Drugs or Substances Associated With Apparent Mineralocorticoid Excess or Renin-Angiotensin System Activation**

- Glucocorticoids and their derivatives
- Glycyrrhizin (licorice)
- Ketoconazole, itraconazole
- Abiraterone acetate
- Synthetic estrogens combined with progestin

Drugs or Substances With Direct Vasopressor Properties

- Immunosuppressive agents (cyclosporine, tacrolimus, and calcineurin inhibitors)
- Recombinant human erythropoietin
- Drugs targeting the vascular endothelial growth factor pathway, including monoclonal antibodies and small-molecule receptor tyrosine kinase inhibitors
- Poly (ADP-ribose) polymerase inhibitors (ovarian cancer)

Drugs or Substances Activating the Sympathetic Nervous System

- Illicit drugs of abuse, such as cocaine and amphetamines
- Epinephrine or phenylephrine derivatives present in over-the-counter oral, nasal, or ophthalmic decongestants
- Ephedrine alkaloids (ephedra or herbal ma-huang)
- Antidepressants, such as serotonin-norepinephrine reuptake inhibitors, monoamine oxidase inhibitors, and tricyclic agents such as amitriptyline
- Appetite suppressants for weight loss (but not incretin drugs)
- Modafinil, a nonamphetamine stimulant

Drugs or Substances With Diverse Mechanisms of Action

- Antiretroviral drugs (lopinavir and ritonavir)
- Nonselective nonsteroidal anti-inflammatory drugs, acetaminophen and selective cyclooxygenase 2 inhibitors
- Bruton tyrosine kinase inhibitors (for chronic lymphocytic leukemia, various lymphomas, etc)
- Teriflunomide (pyrimidine synthesis inhibitor for multiple sclerosis)
- Alcohol

Diuretics

For patients with resistant hypertension and eGFR of 30 mL/min/1.73 m² or greater, 2025 ACC/AHA guidelines recommend switching hydrochlorothiazide to longer-acting thiazide-type diuretics such as chlorthalidone (12.5-25 mg/d) or indapamide (1.25-2.5 mg/d as immediate-release formulation or 1.5 mg/d as slow-release formulation, although slow-release indapamide is not available in the US).^{3,23,24,27} A meta-analysis of 14 RCTs that included 883 patients with office SBP of 140 mm Hg or greater reported that compared with hydrochlorothiazide, SBP was lowered more for both indapamide (mean difference, -5.1 mm Hg [95% CI, -8.7 to -1.6]) and chlorthalidone (mean difference, -3.6 mm Hg [95% CI, -7.3 to 0.0]).⁴³ However, in an RCT (13 523 participants with hypertension, aged ≥65 y and treated with hydrochlorothiazide) reported no significant difference in major cardiovascular outcomes between those randomized to switch to chlorthalidone (12.5 or 25 mg daily) or to continue hydrochlorothiazide (25 or 50 mg daily), over a median follow-up of 2.4 years.⁴⁴

For patients with resistant hypertension and CKD stage 4 or higher, the 2025 AHA/ACC guideline supports using either chlorthalidone with addition of a loop diuretic if further volume control is required or a loop diuretic alone.³ In an RCT of 160 patients with CKD

stage 4 and 24-hour ambulatory blood pressure between 130/80 mm Hg and less than 160/100 mm Hg while receiving at least 1 antihypertensive drug (60% were receiving a loop diuretic), the addition of chlorthalidone (mean dose, 23.1 mg/d) lowered 24-hour ambulatory SBP after 12 weeks of therapy by -11.0 mm Hg (95% CI, -13.9 to -8.1) compared with the placebo response of -0.5 mm Hg (95% CI, -3.5 to 2.5).⁴⁵

The 2025 AHA/ACC guideline recommends rechecking kidney function and serum electrolyte levels 2 to 4 weeks after increasing diuretic doses, switching to loop diuretics or combining thiazide-type diuretics and loop diuretics, or initiating or changing doses of mineralocorticoid receptor antagonists (MRAs).^{3,23,24,27} Hypokalemia from intensified diuretic therapy can be corrected by adding potassium-sparing diuretics (eg, amiloride or triamterene) or MRAs which will further decrease blood pressure (see below).⁴⁶ Potassium supplements can also correct hypokalemia but are less effective at standard doses than potassium-sparing diuretics.^{47,48} An initial and rapid decrease in eGFR over 15 to 30 days in the setting of improved blood pressure control can occur, particularly in individuals taking RAS blockers.^{49,50} A decline in eGFR of less than 30% within 30 days of initiation or escalation of antihypertensive therapy, which occurs due to decreased kidney perfusion pressure associated with improved blood pressure control, does not indicate structural kidney injury⁵¹ and does not require dose reduction or discontinuation of antihypertensive therapy.^{49,52} In contrast, a decline in eGFR of 30% or greater within 30 days of antihypertensive initiation, especially in patients receiving RAS blockers, may suggest the presence of renal artery stenosis.⁵³

Mineralocorticoid Receptor Antagonists

For patients with true resistant hypertension, the addition of MRAs is recommended in the absence of contraindications such as an eGFR less than 30 mL/min/1.73 m² (per European guidelines)²³ or less than 45 mL/min/1.73 m² (per US guidelines)³ or serum potassium concentration greater than 4.5 mmol/L (Table 3).^{3,23,24,27}

Spironolactone

A network meta-analysis of 24 RCTs that included 3485 patients with resistant hypertension and compared treatment options for management of resistant hypertension reported that compared with placebo, spironolactone decreased office SBP by -13.3 mm Hg (95% CI, -17.89 to -8.72 [4 RCTs]) and 24-hour ambulatory SBP by -8.46 mm Hg (95% CI, -12.54 to -4.38 [2 RCTs]).⁵⁴ Another meta-analysis of 12 RCTs that included 1655 patients with resistant hypertension reported that compared with placebo, spironolactone increased serum potassium levels by 0.20 mmol/L (95% CI, 0.05-0.35).⁶¹ The risk of hyperkalemia with MRA therapies is higher among those 65 years or older with higher dietary potassium intake, elevated baseline serum potassium level, higher MRA dose, comorbidities such as CKD, diabetes, heart failure, and use of medications such as RAS blockers, nonsteroidal anti-inflammatory drugs, heparin, trimethoprim-sulfamethoxazole, and β-blockers.⁶² Potassium binders including patiromer⁶³ or sodium zirconium cyclosilicate⁶⁴ can decrease the risk of hyperkalemia with MRAs. In the AMBER trial, among 295 patients with resistant hypertension taking spironolactone who had an eGFR of 25 to 45 mL/min/1.73 m² and baseline serum potassium level less than 5.1 mmol/L, the incidence of hyperkalemia (≥5.5 mmol/L) meeting protocol-specified

Table 2. Commonly Used Antihypertensive Medications in Fixed-Dose Combination Available in the US^a

Antihypertensive class	Medication combination	Dose range
ACE inhibitor in dual combination		
With thiazide-type diuretic	Benazepril + hydrochlorothiazide	10 mg/12.5 mg to 20 mg/25 mg
	Captopril + hydrochlorothiazide	25 mg/15 mg to 50 mg/25 mg
	Enalapril + hydrochlorothiazide	5 mg/12.5 mg to 10 mg/25 mg
	Fosinopril + hydrochlorothiazide	10 mg/12.5 mg to 20 mg/12.5 mg
	Lisinopril + hydrochlorothiazide	10 mg/12.5 mg to 20 mg/25 mg
	Moexipril + hydrochlorothiazide	7.5 mg/12.5 mg to 15 mg/25 mg
	Quinapril + hydrochlorothiazide	10 mg/12.5 mg to 20 mg/25 mg
With calcium channel blocker	Benazepril + amlodipine	10 mg/2.5 mg to 40 mg/10 mg
	Perindopril + amlodipine	3.5 mg/2.5 mg to 14 mg/10 mg
	Trandolapril + verapamil	1 mg/240 mg to 4 mg/240 mg
Angiotensin receptor blocker in dual combination		
With thiazide-type diuretic	Azilsartan + chlorthalidone	40 mg/12.5 mg to 40 mg/25 mg
	Candesartan + hydrochlorothiazide	16 mg/12.5 mg to 32 mg/25 mg
	Irbesartan + hydrochlorothiazide	150 mg/12.5 mg to 300 mg/25 mg
	Losartan + hydrochlorothiazide	50 mg/12.5 mg to 100 mg/25 mg
	Olmesartan + hydrochlorothiazide	20 mg/12.5 mg to 40 mg/25 mg
	Telmisartan + hydrochlorothiazide	40 mg/12.5 mg to 80 mg/25 mg
	Valsartan + hydrochlorothiazide	80 mg/12.5 mg to 320 mg/25 mg
With calcium channel blocker	Olmesartan + amlodipine	20 mg/5 mg to 40 mg/10 mg
	Telmisartan + amlodipine	40 mg/5 mg to 80 mg/10 mg
	Valsartan + amlodipine	160 mg/5 mg/12.5 mg to 320 mg/10 mg/25 mg
With β -blocker	Valsartan + nebivolol	80 mg/5 mg
Angiotensin receptor blocker in triple combination		
With calcium channel blocker + thiazide-type diuretic	Olmesartan + amlodipine + hydrochlorothiazide	20 mg/5 mg/12.5 mg to 40 mg/10 mg/25 mg
	Valsartan + amlodipine + hydrochlorothiazide	160 mg/5 mg/12.5 mg to 320 mg/10 mg/25 mg
	Telmisartan + amlodipine + indapamide	10/1.25/0.625 mg; 20/2.5/1.25 mg; and 40/5/2.5 mg
β-Blocker in dual combination		
With thiazide-type diuretic	Atenolol + chlorthalidone	50 mg/25 mg to 100 mg/25 mg
	Bisoprolol + hydrochlorothiazide	2.5 mg/6.25 mg to 10 mg/6.25 mg
	Metoprolol tartrate + hydrochlorothiazide	50 mg/25 mg to 100 mg/50 mg
Thiazide-type diuretics in dual combination		
With potassium-sparing diuretic	Hydrochlorothiazide + amiloride	50 mg/5 mg
	Hydrochlorothiazide + triamterene	25 mg/37.5 mg to 50 mg/75 mg
With mineralocorticoid receptor antagonist	Hydrochlorothiazide + spironolactone	25 mg/25 mg

Abbreviations: ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker.

^a Adapted from the 2025 American Heart Association/American College of

Cardiology guidelines³; see also <https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm> (accessed September 3, 2025) for more information.

withdrawal of spironolactone over 12 months was lower among those receiving patiomer (7%) vs placebo (23%).⁶³

Spironolactone may cause dose-related antiandrogenic and progestogenic adverse effects such as menstrual irregularities in women (18% taking 50-100 mg daily dose)⁶⁵ and breast tenderness or gynecomastia in men (17%).^{65,66} In the Randomized Aldactone Evaluation Study (RALES), 9.1% of men (603 participants) treated with 25 mg/d spironolactone for heart failure developed gynecomastia, compared with 1.3% in the placebo group (614 participants, $P < .001$ for the comparison).⁶⁷ Among 5485 participants in the Hypertension Detection and Follow-Up Program (HDFP), Stepped Care, 1.6% of hypertensive men discontinued spironolactone due to impotence compared with 4.4% discontinuing chlorthalidone due to impotence over a 5-year follow-up period.⁶⁸ However, spironolactone-

induced impotence is likely underreported, given that erectile dysfunction affects approximately 15% to 20% of the general male population and is nearly twice as prevalent among hypertensive individuals.⁶⁹

Eplerenone

The steroidal MRA eplerenone, which is associated with rates of hyperkalemia similar to those for spironolactone, is more selective for the mineralocorticoid receptor than spironolactone. Because eplerenone has no effect on androgen and progesterone receptors, it is an alternative treatment to spironolactone, especially for male patients with resistant hypertension, with gynecomastia occurring in only 0.5%.⁶⁶ However, eplerenone is short-acting and less potent than spironolactone and is typically dosed at 50 mg daily to 50 mg

Table 3. Antihypertensive Medications for True Resistant Hypertension (Patient Receiving ≥ 3 Antihypertensive Medications)^a

Medication (daily dose)	Mechanism of action	Efficacy (treatment vs control)	Adverse effects
Preferred medications^b			
Spironolactone (25-50 mg)	Mineralocorticoid receptor antagonism	Network meta-analysis in resistant hypertension ⁵⁴ Mean difference in office SBP vs placebo: -13.30 mm Hg (95% CI, -17.89 to -8.72) Mean difference in 24-h ambulatory SBP vs placebo: -8.46 mm Hg (95% CI, -12.54 to -4.38)	Hyperkalemia ($\approx 3\%$ -5%), acute kidney injury (relative risk, ≈ 2), sexual impotence ($\approx 9\%$), gynecomastia (relative risk, ≈ 5), and menstrual irregularities compared with placebo or standard care
Eplerenone (50-100 mg) (in case of antiandrogenic or progestogenic adverse effects of spironolactone)	Mineralocorticoid receptor antagonism	Cochrane database ⁵⁵ Eplerenone has only been evaluated in patients with uncontrolled hypertension Mean difference in office SBP vs placebo: -9.21 mm Hg (95% CI, -11.08 to -7.34) Comparative trial in primary hypertension ⁵⁶ Eplerenone is less potent than spironolactone: mean changes in office SBP for eplerenone (50 mg and 100 mg) twice a day are approximately 50% to 75% of those observed with spironolactone (50 mg) twice a day	Risk of hyperkalemia and acute kidney injury compared with placebo or standard care No antiandrogenic or progestogenic effects
Alternative treatments^c			
Amiloride (10-20 mg) ^d	Epithelial sodium channel blocker	Amiloride noninferior to spironolactone in patients with resistant hypertension ^{57,58}	Hyperkalemia
β -Blockers (eg, bisoprolol [5-10 mg])	Selective β_1 -receptor blockade	Network meta-analysis in resistant hypertension ⁵⁴ Mean difference in office SBP vs placebo: -8.44 mm Hg (95% CI, -15.48 to -1.40) Mean difference in 24-h ambulatory SBP vs placebo: -7.10 mm Hg (95% CI, -18.12 to 3.92)	Fatigue, bradycardia, sexual impotence
α -1 Blockers (eg, doxazosin [4-8 mg])	α_1 -adrenergic blockade	Network meta-analysis in resistant hypertension ⁵⁴ Mean difference in office SBP vs placebo: -7.62 mm Hg (95% CI, -15.86 to 0.62)	Orthostatic hypotension, falls and fractures
Centrally acting sympatholytic medications (eg, clonidine [0.1-0.3 mg] twice daily)	Central α -2 receptor agonist	ReHOT study ⁵⁹ Office blood pressure decreases from baseline to 3 mo: Clonidine: -13.7 mm Hg (95% CI, -18.4 to -9.0) vs spironolactone: -15.1 mm Hg (95% CI, -19.7 to 10.6) Network meta-analysis in resistant hypertension ⁵⁴ Mean difference in office SBP vs placebo: -11.90 mm Hg (95% CI, -23.29 to -0.52) Mean difference in 24-h ambulatory SBP vs placebo: -4.26 mm Hg (95% CI, -12.10 to 3.59)	Somnolence, dry mouth, constipation, orthostatic hypotension, impotence, rebound hypertension ⁶⁰
Aprocritentan (12.5 mg)	Dual endothelin receptor antagonist	PRECISION trial ¹³ Mean difference in unattended office SBP vs placebo: -3.8 mm Hg (97.5% CI, -6.8 to -0.8) Mean difference in 24 h ambulatory SBP: -4.2 mm Hg (95% CI, -6.2 to -2.1)	Edema or fluid retention (9%)

Abbreviations: eGFR, estimated glomerular filtration rate; MRA, mineralocorticoid receptor antagonist; SBP, systolic blood pressure.

^a Preferred medications are an angiotensin-converting enzyme inhibitor or angiotensin receptor blocker, calcium channel blocker, and thiazide-type diuretic.

^b For patient with true resistant hypertension (if eGFR ≥ 45 mL/min/1.73 m² and

serum potassium < 4.5 mmol/L).

^c When mineralocorticoid receptor antagonists are contraindicated or not tolerated, followed by subsequent lines of treatment sequentially added.

^d If eGFR is 45 mL/min/1.73 m² or greater and serum potassium level is less than 4.5 mmol/L. Should not be given with an MRA.

twice daily to achieve antihypertensive effects (mean difference in office SBP, -9.21 mm Hg [95% CI, -11.08 to -7.34 mm Hg] compared with placebo) (Table 3).^{55,56,70} Eplerenone is not approved for hypertension treatment in many European countries.⁶⁶

Other Antihypertensive Medications for Treatment of Resistant Hypertension

For patients who experience adverse effects with MRA, amiloride (10-20 mg daily) may be considered⁵⁷ if eGFR is 45 mL/min/1.73 m²

or greater and serum potassium level is 4.5 mmol/L or less. A prospective, open-label RCT in South Korea (118 patients with resistant hypertension) reported that low-dose amiloride (5-10 mg/d) was noninferior to low-dose spironolactone (12.5-25 mg/d) in lowering home SBP at week 12, with a between-group difference in home SBP of -0.68 mm Hg (90% CI, -3.50 to 2.14) and home SBP control rates (<130 mm Hg) of 66.1% and 55.2%, respectively ($P = .23$).⁵⁸ Other therapeutic options include β -blockers,^{54,71} α -1 blockers,^{54,71} centrally acting antihypertensive medications,^{54,59,60} nondihydropyridine calcium channel blockers, or direct vasodilators, which may be also used as alternatives to spironolactone or added sequentially to a regimen that includes spironolactone. A dual endothelin receptor antagonist (eg, apocritentan [12.5 mg/d]) may also be used for patients with resistant hypertension who are intolerant of MRAs or for whom MRAs are contraindicated, although studies have not directly compared apocritentan with MRAs or other medications for true resistant hypertension (Table 3).³

Direct vasodilators such as hydralazine or minoxidil may cause severe fluid retention requiring increased doses of loop diuretics and reflex sympathetic tachycardia that may need treatment with β -blockers.^{23,27} Apocritentan may cause peripheral edema, fluid retention, and worsening heart failure, particularly in patients with CKD or prior heart failure (Table 3).^{13,72} Apocritentan is contraindicated during pregnancy. The high cost of apocritentan may also limit access.

Renal Denervation

Renal denervation is a minimally invasive catheter-based procedure to treat resistant hypertension despite optimal pharmacologic therapy in patients with eGFR of 40 mL/min/1.73 m² or greater and suitable renal artery anatomy (renal artery diameter of 3-8 mm, length ≥ 20 mm, and no significant renal artery stenosis).^{3,23,24,73,74} The procedure uses a thin dedicated catheter introduced via the femoral artery that is subsequently inserted into the renal arteries to deliver radiofrequency or ultrasound energy to disrupt the sympathetic nerves located within the adventitial layer of the renal artery, thus attenuating both afferent and efferent renal sympathetic nerve activity.^{73,74} This treatment reduces systemic blood pressure through suppression of renin secretion, promotion of natriuresis, and diminution of central sympathetic outflow.^{73,74}

Sham-controlled trials⁷⁵⁻⁷⁸ and a meta-analysis of 10 RCTs (2478 patients; 9 studies required blood pressure of 140/90 mm Hg or greater and 4 explicitly required resistant hypertension)⁷⁹ have demonstrated that renal denervation compared with a sham procedure reduced 24-hour ambulatory SBP by -4.4 mm Hg (95% CI, -6.1 to -2.7) and office SBP by -6.6 mm Hg (95% CI, -9.7 to -3.6). However, only about two-thirds of patients in RCTs responded meaningfully to the procedure with an SBP decrease of 5 mm Hg or more, and reliable predictors of blood pressure response to renal denervation are not currently established.^{73,74} After renal denervation, most patients continue to require antihypertensive medications, although often at lower doses or with fewer medications. Long-term follow-up data show sustained blood pressure lowering associated with renal denervation. A meta-analysis of 6 RCTs and 12 observational studies including 2220 patients with hypertension (including resistant hypertension) demonstrated a reduction from baseline in office SBP of -23.0 mm Hg (95% CI, -26.8 to -19.1 ; 15 studies [2040 participants]) and in 24-hour ambulatory SBP of -13.6 mm Hg (95% CI, -16.5 to -10.8 ; 11 studies [1018 participants]) after a mean

follow-up of 4.4 years (range, 3-9.4).⁸⁰ However, RCTs of renal denervation include highly selected patients who were often at relatively low cardiovascular risk and without comorbidities such as stage 4 CKD, and the renal denervation procedures were performed by physicians with specialized training in the procedure.

Procedural vascular complications at the femoral artery access site were reported in 8 of 1598 patients (0.5%) undergoing renal denervation vs 2 of 1197 (0.17%) in the sham-control group (OR, 1.69 [95% CI, 0.57-5.00]).⁷⁹ A meta-analysis of 50 studies including randomized and nonrandomized trials and registries (5769 patients) reported an annual incidence of renal artery stent implantation to treat renal artery stenosis as a complication following renal denervation using radiofrequency energy of 0.2% per year (95% CI, 0.12%-0.29%).⁸¹

The 2025 AHA/ACA guideline recommends that a detailed evaluation should be performed to exclude secondary causes of hypertension prior to renal denervation and that renal denervation should be considered for patients with resistant hypertension whose blood pressure remains elevated despite optimal antihypertensive therapy, including MRA, or for patients who are intolerant of or have contraindications to MRA.³ In contrast, the European Society of Hypertension guidelines support renal denervation as an option after true resistant hypertension is confirmed.^{3,23,24,73,74} The decision to proceed with renal denervation should involve shared decision-making with the patient and treating clinicians, but access may be limited by its cost and need for specialized expertise.

Prognosis

Patients with resistant hypertension have a higher rates of hypertension-mediated organ damage compared with patients with controlled hypertension of similar duration.⁸² In a large Spanish observational cohort study in which 8146 patients with controlled hypertension and 8577 with resistant hypertension were followed up until death or for 5 to 10 years, true resistant hypertension confirmed by 24-hour ABPM ($\geq 130/80$ mm Hg) was associated with an increased risk of cardiovascular mortality (adjusted HR, 1.68 [95% CI, 1.43-1.98]; absolute risk increase: 10.3% [95% CI, 8.7%-12.1%]) and all-cause mortality (adjusted HR, 1.45 [95% CI, 1.32-1.60]; absolute risk increase, 26.2% [95% CI, 23.5%-28.9%]) over a median follow-up of 9.7 years.⁸³

Investigational Therapies

Investigational treatments include selective inhibitors of aldosterone synthesis such as baxdrostat⁸⁴ or lorundrostat,⁸⁵ which can lower both blood pressure and serum aldosterone in patients with resistant hypertension. In addition, small interfering RNA agents (such as zilebesiran) inhibit the translation of the angiotensinogen messenger RNA, resulting in almost complete suppression of angiotensinogen production by the liver. Angiotensinogen is the precursor to angiotensin I subsequently converted by ACE to angiotensin II, a potent vasoconstrictor.⁸⁶ In a phase 2 RCT, 1491 patients with untreated stage 2 hypertension or uncontrolled hypertension despite taking 1 to 2 antihypertensive medications randomly received open-label indapamide, amlodipine, or olmesartan for 4 weeks. Of these, 663 patients with 24-hour ambulatory SBP of 130 mm Hg or greater and less than 160 mm Hg were rerandomized to a single subcutaneous dose of zilebesiran (600 mg) or placebo added to background therapy.⁸⁷ At 3 months, compared with placebo + each background therapy (indapamide, amlodipine, or olmesartan), 24-hour ambulatory SBP was reduced by -12.1 mm Hg (95% CI, -16.5 to -7.6)

with zilebesiran + indapamide, by -9.7 mm Hg (95% CI, -12.9 to -6.6) with zilebesiran + amlodipine, and by -4.5 mm Hg (95% CI, -8.2 to -0.8) with zilebesiran + olmesartan.⁸⁷

Limitations

This review has limitations. First, some relevant publications may have been missed. Second, a formal quality assessment of the included articles was not performed. Third, few RCTs have exclusively enrolled patients with true resistant hypertension. Fourth, many patients with resistant hypertension have comorbidities that may confound outcomes.

Conclusions

True resistant hypertension affects 10% of patients treated for hypertension and is diagnosed after excluding white-coat hypertension by out-of-office blood pressure measurements, medication non-adherence, and secondary hypertension such as primary aldosteronism. First-line treatment includes lifestyle modifications and diuretic therapy with chlorthalidone and combination tablets of antihypertensives. Spironolactone and renal denervation decrease blood pressure in patients with true resistant hypertension.

ARTICLE INFORMATION

Accepted for Publication: January 29, 2026.

Published Online: March 23, 2026.
doi:10.1001/jama.2026.1221

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Conflict of Interest Disclosures: Dr Azizi reported receiving grants from Novartis, AstraZeneca, Recor, and Sonivie and receiving personal fees from Novartis, AstraZeneca, Recor, Sonivie, Servier, Alnylam, Boehringer Ingelheim, Novo Nordisk, and Medtronic outside the submitted work. Dr Vongpatanasin reported receiving consulting fees from AstraZeneca both during the conduct of the study and outside the submitted work. Dr Fisher reported receiving grants from Recor Medical and receiving consulting fees from Recor Medical, AstraZeneca, Aktia, and Alnylam outside the submitted work. Dr Mahfoud reported receiving personal fees from Ablative Solutions, Medtronic, Merck, and Recor during the conduct of the study; receiving grants from AstraZeneca, Inari, Novartis, and Philips outside the submitted work; receiving support from Deutsche Forschungsgemeinschaft (SFB TRR219, Project-ID 322900939) and Deutsche Herzstiftung, paid to a former employer, Saarland University, which has received scientific support from Ablative Solutions, Medtronic, and Recor Medical; and, until May 2024, receiving speaker honoraria/consulting fees from Ablative Solutions, AstraZeneca, Inari, Medtronic, Merck, Novartis, Philips, and Recor Medical outside the submitted work. Dr Amar reported receiving lecture fees from AstraZeneca outside the submitted work. Dr Kirtane reported receiving research grants to Columbia University and/or Cardiovascular Research Foundation from Abbott

Vascular, Amgen, Biotronik, Boston Scientific, Bolt Medical, CathWorks, Concept Medical, Cordis, Magenta Medical, Medtronic, Neurotronic, Philips, ReCor Medical, and Supira (also includes fees paid to Columbia University and/or Cardiovascular Research Foundation for consulting and/or speaking engagements in which Dr Kirtane controlled the content); holding equity options in Bolt Medical, Airiver, Aeroflow, Lumen Medical, and Tonic Medical; receiving consulting fees from Sonivie; and receiving travel expenses and meals from Amgen, Medtronic, Biotronik, Boston Scientific, Abbott Vascular, CathWorks, Concept Medical, Novartis, Philips, Abiomed, ReCor Medical, Chiesi, Supira, and Shockwave outside the submitted work.

Submissions: We encourage authors to submit papers for consideration as a Review. Please contact Kristin Walter, MD, at kristin.walter@jamanetwork.org.

REFERENCES

1. NCD Risk Factor Collaboration (NCD-RisC). Worldwide trends in hypertension prevalence and progress in treatment and control from 1990 to 2019: a pooled analysis of 1201 population-representative studies with 104 million participants. *Lancet*. 2021;398(10304):957-980. doi:10.1016/S0140-6736(21)01330-1
2. Jaeger BC, Chen L, Foti K, et al. Hypertension statistics for US adults: an open-source web application for analysis and visualization of National Health and Nutrition Examination Survey data. *Hypertension*. 2023;80(6):1311-1320. doi:10.1161/HYPERTENSIONAHA.123.20900
3. Jones DW, Ferdinand KC, Taler SJ, et al. AHA/ACC/AANP/AAHA/ABC/ACCP/ACPM/AGS/AMA/ASPC/NMA/PCNA/SGIM guideline for the prevention, detection, evaluation and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2025;152(11):e114-e218. doi:10.1161/CIR.0000000000001356
4. Akinyelure OP, Sakhuja S, Colvin CL, et al. Cardiovascular health and transition from controlled blood pressure to apparent treatment resistant hypertension: the Jackson Heart Study and the REGARDS Study. *Hypertension*. 2020;76(6):1953-1961. doi:10.1161/HYPERTENSIONAHA.120.15890
5. Noubiap JJ, Nansseu JR, Nyaga UF, Sime PS, Francis I, Bigna JJ. Global prevalence of resistant hypertension: a meta-analysis of data from 3.2 million patients. *Heart*. 2019;105(2):98-105. doi:10.1136/heartjnl-2018-313599
6. Tanner RM, Calhoun DA, Bell EK, et al. Prevalence of apparent treatment-resistant hypertension among individuals with CKD. *Clin J Am Soc Nephrol*. 2013;8(9):1583-1590. doi:10.2215/CJN.00550113
7. Sim JJ, Bhandari SK, Shi J, et al. Characteristics of resistant hypertension in a large, ethnically diverse hypertension population of an integrated health system. *Mayo Clin Proc*. 2013;88(10):1099-1107. doi:10.1016/j.mayocp.2013.06.017
8. Hou H, Zhao Y, Yu W, et al. Association of obstructive sleep apnea with hypertension: a systematic review and meta-analysis. *J Glob Health*. 2018;8(1):010405. doi:10.7189/jogh.08.010405
9. Cecchini M, Filippini T, Whelton PK, et al. Alcohol intake and risk of hypertension: a systematic review and dose-response meta-analysis of nonexperimental cohort studies. *Hypertension*. 2024;81(8):1701-1715. doi:10.1161/HYPERTENSIONAHA.124.22703
10. Piano MR, Marcus GM, Aycock DM, et al; American Heart Association Council on Lifestyle and Cardiometabolic Health; Council on Cardiovascular and Stroke Nursing; Council on Clinical Cardiology; and Stroke Council. Alcohol use and cardiovascular disease: a scientific statement from the American Heart Association. *Circulation*. 2025;152(1):e7-e21. doi:10.1161/CIR.0000000000001341
11. Calhoun DA. Hyperaldosteronism as a common cause of resistant hypertension. *Annu Rev Med*. 2013;64:233-247. doi:10.1146/annurev-med-042711-135929
12. Grassi G, Quarti-Trevano F, Cuspidi C, Sanidas E, Mancia G, Thomopoulos C. Sympathetic overactivation in the resistant hypertensive phenotype: a meta-analysis of published studies. *Hypertension*. 2025;82(8):1346-1354. doi:10.1161/HYPERTENSIONAHA.125.24749
13. Schlaich MP, Bellet M, Weber MA, et al; PRECISION Investigators. Dual endothelin antagonist apocritentan for resistant hypertension (PRECISION): a multicentre, blinded, randomised, parallel-group, phase 3 trial. *Lancet*. 2022;400(10367):1927-1937. doi:10.1016/S0140-6736(22)02034-7
14. Mendes M, Dubourg J, Blanchard A, et al. Copeptin is increased in resistant hypertension. *J Hypertens*. 2016;34(12):2458-2464. doi:10.1097/HJH.0000000000001106
15. Jackson AM, Jhund PS, Anand IS, et al. Sacubitril-valsartan as a treatment for apparent resistant hypertension in patients with heart failure and preserved ejection fraction. *Eur Heart J*. 2021;42(36):3741-3752. doi:10.1093/eurheartj/ehab499

16. Quinaglia T, Martins LC, Figueiredo VN, et al. Non-dipping pattern relates to endothelial dysfunction in patients with uncontrolled resistant hypertension. *J Hum Hypertens*. 2011;25(11):656-664. doi:10.1038/jhh.2011.43
17. Gaddam KK, Nishizaka MK, Pratt-Ubunama MN, et al. Characterization of resistant hypertension: association between resistant hypertension, aldosterone, and persistent intravascular volume expansion. *Arch Intern Med*. 2008;168(11):1159-1164. doi:10.1001/archinte.168.11.1159
18. He FJ, Tan M, Ma Y, MacGregor GA. Salt reduction to prevent hypertension and cardiovascular disease: JACC state-of-the-art review. *J Am Coll Cardiol*. 2020;75(6):632-647. doi:10.1016/j.jacc.2019.11.055
19. Charchar FJ, Prestes PR, Mills C, et al. Lifestyle management of hypertension: International Society of Hypertension position paper endorsed by the World Hypertension League and European Society of Hypertension. *J Hypertens*. 2024;42(1):23-49. doi:10.1097/HJH.0000000000003563
20. Mente A, O'Donnell MJ, Rangarajan S, et al; PURE Investigators. Association of urinary sodium and potassium excretion with blood pressure. *N Engl J Med*. 2014;371(7):601-611. doi:10.1056/NEJMoa1311989
21. Genta-Pereira DC, Pedrosa RP, Lorenzi-Filho G, Drager LF. Sleep disturbances and resistant hypertension: association or causality? *Curr Hypertens Rep*. 2014;16(8):459. doi:10.1007/s11906-014-0459-3
22. White LH, Bradley TD, Logan AG. Pathogenesis of obstructive sleep apnoea in hypertensive patients: role of fluid retention and nocturnal rostral fluid shift. *J Hum Hypertens*. 2015;29(6):342-350. doi:10.1038/jhh.2014.94
23. Mancia G, Kreutz R, Brunström M, et al. 2023 ESH guidelines for the management of arterial hypertension: the Task Force for the Management of Arterial Hypertension of the European Society of Hypertension: Endorsed by the International Society of Hypertension (ISH) and the European Renal Association (ERA). *J Hypertens*. 2023;41(12):1874-2071. doi:10.1097/HJH.0000000000003480
24. McEvoy JW, McCarthy CP, Bruno RM, et al; ESC Scientific Document Group. 2024 ESC guidelines for the management of elevated blood pressure and hypertension. *Eur Heart J*. 2024;45(38):3912-4018. doi:10.1093/eurheartj/ehae178
25. de la Sierra A, Segura J, Banegas JR, et al. Clinical features of 8295 patients with resistant hypertension classified on the basis of ambulatory blood pressure monitoring. *Hypertension*. 2011;57(5):898-902. doi:10.1161/HYPERTENSIONAHA.110.168948
26. Choudhry NK, Kronish IM, Vongpatanasin W, et al; American Heart Association Council on Hypertension; Council on Cardiovascular and Stroke Nursing; and Council on Clinical Cardiology. Medication adherence and blood pressure control: a scientific statement from the American Heart Association. *Hypertension*. 2022;79(1):e1-e14. doi:10.1161/HYP.0000000000000203
27. Carey RM, Calhoun DA, Bakris GL, et al; American Heart Association Professional/Public Education and Publications Committee of the Council on Hypertension; Council on Cardiovascular and Stroke Nursing; Council on Clinical Cardiology; Council on Genomic and Precision Medicine; Council on Peripheral Vascular Disease; Council on Quality of Care and Outcomes Research; and Stroke Council. Resistant hypertension: detection, evaluation, and management: a scientific statement from the American Heart Association. *Hypertension*. 2018;72(5):e53-e90. doi:10.1161/HYP.0000000000000084
28. Brown JM, Siddiqui M, Calhoun DA, et al. The unrecognized prevalence of primary aldosteronism: a cross-sectional study. *Ann Intern Med*. 2020;173(1):10-20. doi:10.7326/M20-0065
29. Adler GK, Stowasser M, Correa RR, et al. Primary aldosteronism: an endocrine society clinical practice guideline. *J Clin Endocrinol Metab*. 2025;110(9):2453-2495. doi:10.1210/clinem/dgaf284
30. Grossman A, Messerli FH, Grossman E. Drug induced hypertension—an unappreciated cause of secondary hypertension. *Eur J Pharmacol*. 2015;763(pt A):15-22. doi:10.1016/j.ejphar.2015.06.027
31. Vitarello JA, Fitzgerald CJ, Cluett JL, Juraschek SP, Anderson TS. Prevalence of medications that may raise blood pressure among adults with hypertension in the United States. *JAMA Intern Med*. 2022;182(1):90-93. doi:10.1001/jamainternmed.2021.6819
32. Blood Pressure Lowering Treatment Trialists' Collaboration. Pharmacological blood pressure lowering for primary and secondary prevention of cardiovascular disease across different levels of blood pressure: an individual participant-level data meta-analysis. *Lancet*. 2021;397(10285):1625-1636. doi:10.1016/S0140-6736(21)00590-0
33. Blood Pressure Lowering Treatment Trialists' Collaboration. Age-stratified and blood-pressure-stratified effects of blood-pressure-lowering pharmacotherapy for the prevention of cardiovascular disease and death: an individual participant-level data meta-analysis. *Lancet*. 2021;398(10305):1053-1064. doi:10.1016/S0140-6736(21)01921-8
34. Wallach JD, Yoon S, Doernberg H, et al. Associations between surrogate markers and clinical outcomes for nononcologic chronic disease treatments. *JAMA*. 2024;331(19):1646-1654. doi:10.1001/jama.2024.4175
35. Yin X, Rodgers A, Perkovic A, et al. Effects of salt substitutes on clinical outcomes: a systematic review and meta-analysis. *Heart*. 2022;108(20):1608-1615. doi:10.1136/heartjnl-2022-321332
36. Neter JE, Stam BE, Kok FJ, Grobbee DE, Geleijnse JM. Influence of weight reduction on blood pressure: a meta-analysis of randomized controlled trials. *Hypertension*. 2003;42(5):878-884. doi:10.1161/01.HYP.0000094221.86888.AE
37. Blumenthal JA, Hinderliter AL, Smith PJ, et al. Effects of lifestyle modification on patients with resistant hypertension: results of the TRIUMPH randomized clinical trial. *Circulation*. 2021;144(15):1212-1226. doi:10.1161/CIRCULATIONAHA.121.055329
38. Lopes S, Mesquita-Bastos J, Garcia C, et al. Effect of exercise training on ambulatory blood pressure among patients with resistant hypertension: a randomized clinical trial. *JAMA Cardiol*. 2021;6(11):1317-1323. doi:10.1001/jamacardio.2021.2735
39. Roerecke M, Kaczorowski J, Tobe SW, Gmel G, Hasan OSM, Rehm J. The effect of a reduction in alcohol consumption on blood pressure: a systematic review and meta-analysis. *Lancet*. 2017;2(2):e108-e120. doi:10.1016/S2468-2667(17)30003-8
40. Lastra AC, Neborak JM, Mokheles B. Diagnosis and treatment of obstructive sleep apnea. *JAMA Intern Med*. 2025. doi:10.1001/jamainternmed.2025.2318
41. Sun L, Chang YF, Wang YF, et al. Effect of continuous positive airway pressure on blood pressure in patients with resistant hypertension and obstructive sleep apnea: an updated meta-analysis. *Curr Hypertens Rep*. 2024;26(5):201-211. doi:10.1007/s11906-024-01294-4
42. Parati G, Kjeldsen S, Coca A, Cushman WC, Wang J. Adherence to single-pill versus free-equivalent combination therapy in hypertension: a systematic review and meta-analysis. *Hypertension*. 2021;77(2):692-705. doi:10.1161/HYPERTENSIONAHA.120.15781
43. Roush GC, Ernst ME, Kostis JB, Tandon S, Sica DA. Head-to-head comparisons of hydrochlorothiazide with indapamide and chlorthalidone: antihypertensive and metabolic effects. *Hypertension*. 2015;65(5):1041-1046. doi:10.1161/HYPERTENSIONAHA.114.05021
44. Ishani A, Cushman WC, Leatherman SM, et al; Diuretic Comparison Project Writing Group. Chlorthalidone vs hydrochlorothiazide for hypertension-cardiovascular events. *N Engl J Med*. 2022;387(26):2401-2410. doi:10.1056/NEJMoa2212270
45. Agarwal R, Sinha AD, Cramer AE, et al. Chlorthalidone for hypertension in advanced chronic kidney disease. *N Engl J Med*. 2021;385(27):2507-2519. doi:10.1056/NEJMoa2110730
46. Roush GC, Ernst ME, Kostis JB, Yeasmin S, Sica DA. Dose doubling, relative potency, and dose equivalence of potassium-sparing diuretics affecting blood pressure and serum potassium: systematic review and meta-analyses. *J Hypertens*. 2016;34(1):11-19. doi:10.1097/HJH.0000000000000762
47. Morgan DB, Davidson C. Hypokalaemia and diuretics: an analysis of publications. *BMJ*. 1980;280(6218):905-908. doi:10.1136/bmj.280.6218.905
48. Jackson PR, Ramsay LE, Wakefield V. Relative potency of spironolactone, triamterene and potassium chloride in thiazide-induced hypokalaemia. *Br J Clin Pharmacol*. 1982;14(2):257-263. doi:10.1111/j.1365-2125.1982.tb01970.x
49. Palmer BF. Renal dysfunction complicating the treatment of hypertension. *N Engl J Med*. 2002;347(16):1256-1261. doi:10.1056/NEJMra020676
50. Palmer BF. Angiotensin-converting enzyme inhibitors and angiotensin receptor blockers: what to do if the serum creatinine and/or serum potassium concentration rises. *Nephrol Dial Transplant*. 2003;18(10):1973-1975. doi:10.1093/ndt/gfg282
51. Chen DC, McCallum W, Sarnak MJ, Ku E. Intensive BP control and eGFR declines: are these events due to hemodynamic effects and are changes reversible? *Curr Cardiol Rep*. 2020;22(10):117. doi:10.1007/s11886-020-01365-3
52. Beddhu S, Whelton PK. Early decline in kidney function with intensive systolic blood pressure lowering: does it matter? *Hypertension*. 2020;75(5):1163-1164. doi:10.1161/HYPERTENSIONAHA.120.14784

53. Sarafidis PA, Theodorakopoulou M, Ortiz A, et al. Atherosclerotic renovascular disease: a clinical practice document by the European Renal Best Practice (ERBP) board of the European Renal Association (ERA) and the Working Group Hypertension and the Kidney of the European Society of Hypertension (ESH). *Nephrol Dial Transplant*. 2023;38(12):2835-2850. doi:10.1093/ndt/gfad095
54. Tian Z, Vollmer Barbosa C, Lang H, Bauersachs J, Melk A, Schmidt BMW. Efficacy of pharmacological and interventional treatment for resistant hypertension: a network meta-analysis. *Cardiovasc Res*. 2024;120(1):108-119. doi:10.1093/cvr/cvad165
55. Tam TS, Wu MH, Masson SC, et al. Eplerenone for hypertension. *Cochrane Database Syst Rev*. 2017;2(2):CD008996.
56. Weinberger MH, Roniker B, Krause SL, Weiss RJ. Eplerenone, a selective aldosterone blocker, in mild-to-moderate hypertension. *Am J Hypertens*. 2002;15(8):709-716. doi:10.1016/S0895-7061(02)02957-6
57. Williams B, MacDonald TM, Morant SV, et al; British Hypertension Society Programme of Prevention And Treatment of Hypertension With Algorithm-based Therapy (PATHWAY) Study Group. Endocrine and haemodynamic changes in resistant hypertension, and blood pressure responses to spironolactone or amiloride: the PATHWAY-2 mechanisms substudies. *Lancet Diabetes Endocrinol*. 2018;6(6):464-475. doi:10.1016/S2213-8587(18)30071-8
58. Lee CJ, Ihm SH, Shin DH, et al. Spironolactone vs amiloride for resistant hypertension: a randomized clinical trial. *JAMA*. 2025;333(23):2073-2082. doi:10.1001/jama.2025.5129
59. Krieger EM, Drager LF, Giorgi DMA, et al; ReHOT Investigators. Spironolactone versus clonidine as a fourth-drug therapy for resistant hypertension: the ReHOT randomized study (Resistant Hypertension Optimal Treatment). *Hypertension*. 2018;71(4):681-690. doi:10.1161/HYPERTENSIONAHA.117.10662
60. Érszegi A, Viola R, Bahar MA, et al. Not first-line antihypertensive agents, but still effective—the efficacy and safety of imidazoline receptor agonists: a network meta-analysis. *Pharmacol Res Perspect*. 2024;12(3):e1215. doi:10.1002/prp2.1215
61. Chen C, Zhu XY, Li D, Lin Q, Zhou K. Clinical efficacy and safety of spironolactone in patients with resistant hypertension: a systematic review and meta-analysis. *Medicine (Baltimore)*. 2020;99(34):e21694. doi:10.1097/MD.00000000000021694
62. Palmer BF, Clegg DJ. Hyperkalemia treatment standard. *Nephrol Dial Transplant*. 2024;39(7):1097-1104. doi:10.1093/ndt/gfae056
63. Agarwal R, Rossignol P, Romero A, et al. Patiromer versus placebo to enable spironolactone use in patients with resistant hypertension and chronic kidney disease (AMBER): a phase 2, randomised, double-blind, placebo-controlled trial. *Lancet*. 2019;394(10208):1540-1550. doi:10.1016/S0140-6736(19)32135-X
64. Packham DK, Rasmussen HS, Lavin PT, et al. Sodium zirconium cyclosilicate in hyperkalemia. *N Engl J Med*. 2015;372(3):222-231. doi:10.1056/NEJMoa1411487
65. Kim GK, Del Rosso JQ. Oral spironolactone in post-teenage female patients with acne vulgaris: practical considerations for the clinician based on current data and clinical experience. *J Clin Aesthet Dermatol*. 2012;5(3):37-50.
66. Ménard J. The 45-year story of the development of an anti-aldosterone more specific than spironolactone. *Mol Cell Endocrinol*. 2004;217(1-2):45-52. doi:10.1016/j.mce.2003.10.008
67. Pitt B, Zannad F, Remme WJ, et al; Randomized Aldactone Evaluation Study Investigators. The effect of spironolactone on morbidity and mortality in patients with severe heart failure. *N Engl J Med*. 1999;341(10):709-717. doi:10.1056/NEJM199909023411001
68. Curb JD, Borhani NO, Blaszkowski TP, Zimbaldi N, Fotiu S, Williams W. Long-term surveillance for adverse effects of antihypertensive drugs. *JAMA*. 1985;253(22):3263-3268. doi:10.1001/jama.1985.03350460063022
69. Viigimaa M, Vlachopoulos C, Doumas M, et al; European Society of Hypertension Working Group on Sexual Dysfunction. Update of the position paper on arterial hypertension and erectile dysfunction. *J Hypertens*. 2020;38(7):1220-1234. doi:10.1097/HJH.0000000000002382
70. Colussi G, Catena C, Sechi LA. Spironolactone, eplerenone and the new aldosterone blockers in endocrine and primary hypertension. *J Hypertens*. 2013;31(1):3-15. doi:10.1097/HJH.0b013e3283599b6a
71. Williams B, MacDonald TM, Morant S, et al; British Hypertension Society PATHWAY Studies Group. Spironolactone versus placebo, bisoprolol, and doxazosin to determine the optimal treatment for drug-resistant hypertension (PATHWAY-2): a randomised, double-blind, crossover trial. *Lancet*. 2015;386(10008):2059-2068. doi:10.1016/S0140-6736(15)00257-3
72. Flack JM, Schlaich MP, Weber MA, et al. Aprocritentan for blood pressure reduction in Black patients. *Hypertension*. 2025;82(4):601-610. doi:10.1161/HYPERTENSIONAHA.124.24142
73. Cluett JL, Blazek O, Brown AL, et al; American Heart Association Council on Hypertension; Council on Cardiovascular and Stroke Nursing; Council on the Kidney in Cardiovascular Disease; and Council on Peripheral Vascular Disease. Renal denervation for the treatment of hypertension: a scientific statement from the American Heart Association. *Hypertension*. 2024;81(10):e135-e148. doi:10.1161/HYP.0000000000000240
74. Barbato E, Azizi M, Schmieder RE, et al. Renal denervation in the management of hypertension in adults: a clinical consensus statement of the ESC Council on Hypertension and the European Association of Percutaneous Cardiovascular Interventions (EAPCI). *Eur Heart J*. 2023;44(15):1313-1330. doi:10.1093/eurheartj/ehad054
75. Azizi M, Sanghvi K, Saxena M, et al; RADIANCE-HTN Investigators. Ultrasound renal denervation for hypertension resistant to a triple medication pill (RADIANCE-HTN TRIO): a randomised, multicentre, single-blind, sham-controlled trial. *Lancet*. 2021;397(10293):2476-2486. doi:10.1016/S0140-6736(21)00788-1
76. Kandzari DE, Townsend RR, Kario K, et al; SPYRAL HTN-ON MED Investigators. Safety and efficacy of renal denervation in patients taking antihypertensive medications. *J Am Coll Cardiol*. 2023;82(19):1809-1823. doi:10.1016/j.jacc.2023.08.045
77. Jiang X, Mahfoud F, Li W, et al. Efficacy and safety of catheter-based radiofrequency renal denervation in Chinese patients with uncontrolled hypertension: the randomized, sham-controlled, multi-center Iberis-HTN trial. *Circulation*. 2024;150(20):1588-1598. doi:10.1161/CIRCULATIONAHA.124.069215
78. Azizi M, Mahfoud F, Weber MA, et al; RADIANCE-HTN Investigators. Effects of renal denervation vs sham in resistant hypertension after medication escalation: prespecified analysis at 6 months of the RADIANCE-HTN TRIO randomized clinical trial. *JAMA Cardiol*. 2022;7(12):1244-1252. doi:10.1001/jamacardio.2022.3904
79. Vukadinović D, Lauder L, Kandzari DE, et al. Effects of catheter-based renal denervation in hypertension: a systematic review and meta-analysis. *Circulation*. 2024;150(20):1599-1611. doi:10.1161/CIRCULATIONAHA.124.069709
80. Mahfoud F, Kandzari DE, Schlaich MP, et al. Long-term outcomes following radiofrequency renal denervation: meta-analysis of 18 reports. *Eur J Prev Cardiol*. 2025;zwaf368. doi:10.1093/eurjpc/zwaf368
81. Townsend RR, Walton A, Hettrick DA, et al. Review and meta-analysis of renal artery damage following percutaneous renal denervation with radiofrequency renal artery ablation. *EuroIntervention*. 2020;16(1):89-96. doi:10.4244/EIJ-D-19-00902
82. Cuspidi C, Macca G, Sampieri L, et al. High prevalence of cardiac and extracardiac target organ damage in refractory hypertension. *J Hypertens*. 2001;19(11):2063-2070. doi:10.1097/00004872-200111000-00018
83. de la Sierra A, Ruilope LM, Staplin N, et al. Resistant hypertension and mortality: an observational cohort study. *Hypertension*. 2024;81(11):2350-2356. doi:10.1161/HYPERTENSIONAHA.124.23276
84. Flack JM, Azizi M, Brown JM, et al; BaxHTN Investigators. Efficacy and safety of baxdrostat in uncontrolled and resistant hypertension. *N Engl J Med*. 2025;393(14):1363-1374. doi:10.1056/NEJMoa2507109
85. Laffin LJ, Kopjar B, Melgaard C, et al; Advance-HTN Investigators. Lorundrostat efficacy and safety in patients with uncontrolled hypertension. *N Engl J Med*. 2025;392(18):1813-1823. doi:10.1056/NEJMoa2501440
86. Ye D, Cruz-López EO, Tu HC, Zlatev I, Danser AHJ. Targeting angiotensinogen with N-acetylgalactosamine-conjugated small interfering RNA to reduce blood pressure. *Arterioscler Thromb Vasc Biol*. 2023;43(12):2256-2264. doi:10.1161/ATVBAHA.123.319897
87. Desai AS, Karns AD, Badariene J, et al; KARDIA-2 Study Group. Add-on treatment with zilebesiran for inadequately controlled hypertension: the KARDIA-2 randomized clinical trial. *JAMA*. 2025;334(1):46-55. doi:10.1001/jama.2025.6681